



Republic of the Philippines
Isabela State University
Echague, Isabela



College of Computing Studies, Information and Communication Technology

WEEK 1:

Project Review and Development Planning



FORM 1

PANEL RECOMMENDATION COMPLIANCE MATRIX

No.	Panel Recommendation	Planned Action	Evidence	Status
1	Clarify how the system addresses the issue of delayed field data reception.	Expanded the background section to explicitly state how the offline-syncing mobile app and the automated analytics dashboard resolve data reception bottlenecks.	pg. 3	Completed
2	Clarify the testing mechanism for the weather analytics module and spraying recommendations.	Removed the weather API and spraying recommendations from the objectives, as there is no available domain expert to validate these specific features but remained as a weather for visualization guide.	pg. 4	Completed
3	Simplify the problem statement and structure the objectives in ascending thematic order.	Reworded and reorganized the project objectives in ascending thematic order to clearly establish the progression from the centralized web portal to the offline-capable mobile application.	pg. 5	Completed
4	Add appropriate citation to support the chosen RAD methodology.	Integrated relevant academic citations to support and justify the use of the Rapid Application Development (RAD) methodology.	pg. 31	Completed
5	Rephrase the introduction of the diagram.	Revised the introduction to the System Architecture diagram to provide a clearer, more concise overview of the dual-platform integration.	pg. 45	Completed



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6	Use noun-based labels to represent data flow in the Context Diagram and Data Flow Diagrams.	Updated all flow labels across the Context Diagram and DFDs to strictly use noun phrases instead of verbs, aligning with standard diagramming conventions.	pgs. 76-83	Completed
7	Simplify the Level 1 (Admin) and Level 2 (Field Technician and Farmer) Data Flow Diagrams.	De-cluttered and simplified the Level 1 and Level 2 DFDs by consolidating redundant processes and clarifying data stores.	pg. 78	Completed
8	Refine the system scope to include only the most essential features.	Refined the project scope to emphasize its primary function to manage subsidies and support programs and as a knowledge access portal.	pgs. 1-8	Completed



FORM 2
PROJECT SCOPE VALIDATION FORM

A. Objectives Validation

Objective	Still Applicable?	Remarks
Develop a centralized web-based portal that digitizes farmer records, manage subsidies and support programs and generates automated reports through an analytics dashboard.	Yes	This aligns with the primary goal of improving the management of farmers' subsidies and support programs.
Integrate an offline-capable mobile application for field technicians with QR code verification and GPS-based data collection for monitoring crops.	Yes	This resolves the bottleneck of decentralized workflows and manual ID verification at distribution points
Implement an SMS advisory to the system that sends alerts, farming recommendations, and distribution schedules to farmers.	Yes	This addresses the communication gap for vulnerable farmers lacking internet access.
Develop a module to generate farmer ID cards with unique QR codes for easy verification and identification of priority groups.	Yes	This ensures the systematic prioritization of vulnerable groups like seniors and PWDs during distributions.
Design and implement predictive analytics in the dashboard that provides data-driven forecasts on crop yields, and agricultural risks to support proactive planning and decision-making.	Yes	This transitions the MAO from a reactive basis to a proactive, data-driven institution.
Evaluate the system quality based on ISO/IEC 25010 standards in terms of functionality, performance efficiency, usability, reliability, and security.	Yes	This ensures the finalized system meets strict software quality standards for reliable and secure resource distribution.



B. Requirements Validation

Requirement	Validated	Remarks
Functional Requirements	☒	Core functions validated include RSBSA-aligned database management, automated statutory report generation, SMS notifications, and offline-capable QR scanning.
Non-Functional Requirements	☒	System must be evaluated against strict ISO/IEC 25010 quality standards for reliability, security, and performance.
User Requirements	☒	Roles are clearly defined for MAO administrators (web portal access), field technicians (mobile app access), and farmers (passive interaction via SMS and QR IDs).
Hardware Requirements	☒	The study explicitly excludes the physical procurement or provision of hardware, such as smartphones or local office servers.
Software Requirements	☒	Validated as a dual-platform digital solution comprising a Web-Based Administrative Portal and a Mobile Application.

C. Scope Changes

List proposed modifications.

Proposed Change	Reason	Adviser Approval
Removed the weather API and spraying recommendations from the objectives.	There is no available domain expert to validate these specific features.	<u>Renalyn G. Tecson, DIT</u>
Refined the project scope to emphasize its primary function to manage subsidies and support programs and as a knowledge access portal.	Based on panel recommendations to include only the most essential system features.	<u>Renalyn G. Tecson, DIT</u>

FORM 3
WORK BREAKDOWN STRUCTURE (WBS)

Major Deliverable	Task	Assigned Member	Due Week
Front-End Development	Scaffold Ionic Vue codebase (PWA & Android)	John Mitchel Dupitas / Justin Iddurut	
	Setup Vue Router (Role-based access)	Justin Iddurut	
	Implement offline queue/Pinia state	Justin Iddurut	
	QR-based farmer verification UI	Justin Iddurut	
	Scaffold Laravel REST API & Sanctum Auth	John Mitchel Dupitas	
Back-End Development	Standardize JSON API response structures	John Mitchel Dupitas	
	Subsidy distribution tracking endpoints	John Mitchel Dupitas	
	Translate ERD to MariaDB schema	John Mitchel Dupitas	
Database Development	Implement UUID generation & soft deletes	John Mitchel Dupitas	
	Configure database seeders	John Mitchel Dupitas	
	Test duplicate claim prevention (Constraints)	Dave Raphael Ignacio	
Testing	Test offline-first synchronization	Dave Raphael Ignacio	
	Deploy backend to Railway/Vercel	John Mitchel Dupitas	
Deployment	API Contracts & Architecture Documentation	Entire Team	



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Team Verification

All members must sign below:

1. Dupitas, John Mitchel M.

Signature: JD

Date: July 21, 2026

2. Iddurut, Justin R.

Signature: Jiddurut

Date: July 21, 2026

3. Ignacio, Dave Raphael M.

Signature: De- Ignacio

Date: July 21, 2026



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FORM 4

GANTT CHART SUBMISSION FORM

Activities	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15	W 16	W 17	W 18
Develop ment																		
Testing																		
Deploym ent																		
Evaluatio n																		
Documen tation																		

Milestone Schedule

Milestone	Target Week
Alpha Version	Week 5
Prototype Presentation	Week 7
Beta Version	Week 11
Agency Immersion	Week 12
Deployment	Week 12
Evaluation	Week 16
Mock Defense	Week 17
Final Defense	Week 18



FORM 5
PROJECT DEVELOPMENT PLAN

A. Team Roles

Team Member	Role	Responsibilities
John Mitchel M. Dupitas	Lead Full-Stack Architect	Manage Laravel backend, MariaDB database, deployment, and core Ionic Vue structure.
Justin R. Iddurut	Front-End Developer	Develop Ionic Vue interfaces, QR scanning integration, and mobile UI rendering.
Dave Raphael M. Ignacio	Quality Assuror	Design and execute test suites for ISO/IEC 25010 quality evaluation; perform automated and manual testing on offline-first queue synchronization, manage User Acceptance Testing (UAT) during MAO agency immersion.

B. Development Methodology

- ☐ Agile
- ☐ Scrum
- ☐ Waterfall
- ☐ Incremental
- ☒ Rapid Application Development

B. Development Tools

Category	Tool
Programming Language	PHP , TypeScript / JavaScript
Framework	Laravel (Backend) , Ionic Vue + Capacitor (Frontend)
Database	MariaDB
IDE	Visual Studio Code
Repository	GitHub



D. Risk Management Plan

Risk	Probability	Impact	Mitigation
Primary Key Collisions when syncing offline data.	High	High	Use UUIDs instead of auto-incrementing IDs for all core tables.
Duplicate Subsidy Claims (Double-dipping).	Medium	High	Enforce a Composite Unique Constraint at the database level.
Accidental Deletion of government RSBSA records.	Low	High	Implement <code>\$table->softDeletes()</code> in migrations so records are never permanently erased.

E. Communication Plan

Activity	Schedule
Team Meeting	Twice weekly (Every Tuesday and Friday evening for sprint review and task syncing)
Adviser Consultation	Weekly (Every Thursday / Prior to major milestone submissions)
Client Coordination	Bi-weekly (With Echague MAO focal persons for requirements check and pilot testing updates)



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FORM 6
REPOSITORY SETUP REPORT

Repository Information

Repository Name: agri-akap-frontend , agri-akap-backend

Repository URL:

<https://github.com/JohnMitchelDupitas/agri-akap-frontend.git>

<https://github.com/JohnMitchelDupitas/agri-akap-backend.git>

Member Access

Member	Access Granted
John Mitchel M. Dupitas	Yes
Justin R. Iddurut	Yes
Dave Raphael M. Ignacio	Yes



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☒ Team Members

☐ Select all

Type ▾

☐	 dabe daberphl-10 • Collaborator	
☐	 Iddurut, Justin R. IDDUJUSTIN • Collaborator	
☐	 John Mitchel Dupitas johnmitchelmdupitas • Collaborator	



☒ Folder Structure

JohnMitchelDupitas removed poic and added more reports needed		32e4541 yesterday	🕒 13 Commits
app	removed poic and added more reports needed	yesterday	
bootstrap	firstdraftB	last week	
config	fix bugs	last week	
database	removed poic and added more reports needed	yesterday	
public	initial commit	2 months ago	
resources	initial commit	2 months ago	
routes	removed poic and added more reports needed	yesterday	
storage	initial commit	2 months ago	
tests	initial commit	2 months ago	
.editorconfig	initial commit	2 months ago	
.env.example	firstdraftB	last week	
.gitattributes	initial commit	2 months ago	
.gitignore	initial commit	2 months ago	
.npmrc	initial commit	2 months ago	
README.md	initial commit	2 months ago	
artisan	initial commit	2 months ago	
composer.json	add auth	last month	
composer.lock	add auth	last month	
package-lock.json	add auth	last month	
package.json	add auth	last month	
phpunit.xml	initial commit	2 months ago	
vite.config.js	initial commit	2 months ago	



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JohnMitchelDupitas added mobile features pest and disaster		3a95851 · 20 hours ago	🕒 21 Commits
📁 .vscode	Registration Form		last month
📁 public	Initial commit		3 months ago
📁 src	added mobile features pest and disaster		20 hours ago
📁 tests	Initial commit		3 months ago
📄 .browserslistrc	Initial commit		3 months ago
📄 .env	gira monitort huhu		last month
📄 .eslintignore	Initial commit		3 months ago
📄 .eslintrc.cjs	added sidebar & utils		last month
📄 .gitignore	Initial commit		3 months ago
📄 capacitor.config.ts	firstdraftF		last week
📄 cypress.config.ts	Initial commit		3 months ago
📄 index.html	Initial commit		3 months ago
📄 ionic.config.json	Initial commit		3 months ago
📄 package-lock.json	firstdraftF		last week
📄 package.json	firstdraftF		last week
📄 tsconfig.json	Initial commit		3 months ago
📄 tsconfig.node.json	Initial commit		3 months ago
📄 vite.config.ts	Initial commit		3 months ago

☑ Initial Commit

Commits on Apr 20, 2026	
Initial commit JohnMitchelDupitas committed on Apr 20	0a46517 🔗 ↗
Commits on May 26, 2026	
Initial commit JohnMitchelDupitas committed on May 26	ee5f96c 🔗 ↗



Repository Structure

List folders created.

Folder Name	Purpose
/src (Frontend)	Vue components, views, router, and Pinia stores.
/public (Frontend)	Static assets for the Ionic application
/app (Backend)	Core Laravel application logic, models, and controllers
/database (Backend)	Laravel migrations (farmers, programs, distributions) and seeders.
/routes (Backend)	Sanctum API endpoint definitions and route management.



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ACTIVITY 7 Adviser Consultation

Purpose

To validate project readiness before development starts.

Procedure

1. Present all Week 1 outputs.
2. Discuss project schedule.
3. Discuss risks and responsibilities.
4. Record adviser recommendations.
5. Complete Form 7.

FORM 7 ADVISER CONSULTATION RECORD

Date	Topics Discussed	Adviser Recommendations	Action Items
July 21, 2026	Gantt Chart scheduling and academic timeline alignment.	Accelerate the initial development phase to ensure an Alpha prototype is ready for presentation between the 5th and 6th week.	Front-load the Work Breakdown Structure and update the Gantt Chart to reflect the new Week 6 Prototype milestone.
July 24, 2026	Project Scope Validation and panel compliance.	Proceed with the refined scope, permanently excluding spraying recommendations as agreed upon.	Finalize Chapter 1 revisions and begin backend database schema translation for the core subsidy management features.
July 27, 2026	Risk Management and Repository Setup.	Strictly implement the composite unique constraints and UUIDs in the database to prevent subsidy double-claiming during offline syncing.	Ensure all team members have committed their initial files and grant the adviser viewing access to both agri-akap-frontend and agri-akap-backend repositories.



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Consultation Checklist

Item	Discussed
Project Scope	/
Timeline	/
Team Responsibilities	/
Development Tools	/
Risks	/
Repository Setup	/

Adviser Approval

Adviser Name: Renalyn C. Teson, DIT

Signature: [Signature]

Date: July 24 2026